

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied Business Analytics

Practical – 9

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9. Hypothesis Testing:

a. Formulate null and alternative hypotheses related to a business problem, conduct a hypothesis test using appropriate statistical tests, and interpret the results.

```
import pandas as pd
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt
df = pd.read_csv("supermarket_sales.csv")
```

2. Display the first five records:

```
print("First five records:")
print(df.head())
```

First five records:

	Invoice ID	Branch	City	Customer type	Gender	\
0	750-67-8428	A	Yangon	Member	Female	
1	226-31-3881	C	Naypyitaw	Normal	Female	
2	631-41-3108	A	Yangon	Normal	Male	
3	123-19-1176	A	Yangon	Member	Male	
4	373-73-7910	A	Yangon	Normal	Male	

	Product line	Unit price	Quantity	Tax 5%	Total	Date	\
0	Health and beauty	74.69	7	26.1415	548.9715	1/5/2019	
1	Electronic accessories	15.28	5	3.8200	80.2200	3/8/2019	
2	Home and lifestyle	46.33	7	16.2155	340.5255	3/3/2019	
3	Health and beauty	58.22	8	23.2880	489.0480	1/27/2019	
4	Sports and travel	86.31	7	30.2085	634.3785	2/8/2019	

	Time	Payment	cogs	gross margin percentage	gross income	Rating
0	13:08	Ewallet	522.83	4.761905	26.1415	9.1
1	10:29	Cash	76.40	4.761905	3.8200	9.6
2	13:23	Credit card	324.31	4.761905	16.2155	7.4
3	20:33	Ewallet	465.76	4.761905	23.2880	8.4
4	10:37	Ewallet	604.17	4.761905	30.2085	5.3

3. Select the revenue column:

5. Calculate the mean revenue:

```
mean_revenue = revenue.mean()
print("\nMean Revenue:", mean_revenue)
```

Mean Revenue: 322.966749

6. Assume that the expected average revenue is ₹1,500:

```
expected_mean = 1500
```

7. Write the null hypothesis: $H_0: \mu = 1500$

8. Write the alternative hypothesis: $H_1: \mu \neq 1500$

```
revenue = df["Total"]
print("\nRevenue:")
print(revenue)
```

```
Revenue:
0      548.9715
1       80.2200
2      340.5255
3      489.0480
4      634.3785
...
995     42.3675
996    1022.4900
997     33.4320
998     69.1110
999     649.2990
Name: Total, Length: 1000, dtype: float64
```

4. Check for missing values:

```
print("\nMissing values in revenue:")
print(revenue.isnull().sum())
# Remove missing values if present
revenue = revenue.dropna()
```

Missing values in revenue:

```
# 7. Alternative Hypothesis
# H1:  $\mu_A \neq \mu_B$ 

# 8. Significance level
alpha = 0.05

✓ 0.0s
```

9. Set the significance level as 5%:

```
alpha = 0.05
```

10. Perform a one-sample t-test:

```
t_stat, p_value = stats.ttest_1samp(
    revenue,
    expected_mean
)
```

11. Calculate the p-value:

```
print("\nT-statistic:", t_stat)
print("P-value:", p_value)
```

```
T-statistic: -151.37567896776733
P-value: 0.0
```

12 & 13. Compare p-value with 0.05:**Conclusion:**

The test shows that the average revenue is significantly different from ₹1,500. Therefore, the business should not assume that its average revenue is ₹1,500 based on the given data.

b. Perform A/B testing on a website or marketing campaign to evaluate the effectiveness of different strategies and make data-driven decisions

```
import pandas as pd
from statsmodels.stats.proportion import
proportions_ztest
import matplotlib.pyplot as plt
```

1. Load the dataset

```
data = pd.read_csv("ab_testing.csv")
```

Convert Conversion column into numeric values
If values are 0 and 1 stored as text, this will convert them

```
data["Conversion"] =
pd.to_numeric(data["Conversion"], errors="coerce")
```

2. Display the data

```
print("A/B Testing Dataset:")
print(data)
```

A/B Testing Dataset:			
	User	Group	Conversion
0	14292	B	NaN
1	11682	A	NaN
2	19825	A	NaN
3	16080	B	NaN
4	18851	A	NaN
...	...	...	...
4995	16360	B	NaN
4996	18084	B	NaN
4997	12063	A	NaN
4998	18647	B	NaN
4999	16686	A	NaN

Check for missing values

```
data = data.dropna(subset=["Conversion"])
```

3. Calculate conversion rate for Group A

```
group_A = data[data["Group"] == "A"]
users_A = len(group_A)
conversions_A = group_A["Conversion"].sum()
rate_A = conversions_A / users_A
print("\nGroup A Conversion Rate:", rate_A)
print("Group A Conversion Rate:", rate_A * 100, "%")
```

```
if p_value < alpha:
```

```
    print("Reject the Null Hypothesis")
```

```
else:
```

```
    print("Do not Reject the Null Hypothesis")
```

```
Reject the Null Hypothesis
```

```
Group A Conversion Rate: nan
Group A Conversion Rate: nan %
C:\Users\itlab\AppData\Local\Temp\ipykernel_...
rate_A = conversions_A / users_A
```

4. Calculate conversion rate for Group B

```
group_B = data[data["Group"] == "B"]
users_B = len(group_B)
conversions_B = group_B["Conversion"].sum()
rate_B = conversions_B / users_B
print("\nGroup B Conversion Rate:", rate_B)
print("Group B Conversion Rate:", rate_B * 100, "%")
```

```
Group B Conversion Rate: nan
Group B Conversion Rate: nan %
C:\Users\itlab\AppData\Local\Temp\ipykernel_...
rate_B = conversions_B / users_B
```

5. Compare conversion rates

```
print("\nComparison:")
print("Group A:", rate_A * 100, "%")
print("Group B:", rate_B * 100, "%")
```

```
if rate_A > rate_B:
```

```
    print("Group A has a higher conversion rate.")
```

```
elif rate_B > rate_A:
```

```
    print("Group B has a higher conversion rate.")
```

```
else:
```

```
    print("Both groups have the same conversion rate.")
```

```
Comparison:
Group A: nan %
Group B: nan %
Both groups have the same conversion rate.
```

6. Null Hypothesis

```
print("\nH0: pA = pB")
```

H0: $p_A = p_B$

7. Alternative Hypothesis

`print("H1:  $p_A \neq p_B$ ")`

H1: $p_A \neq p_B$

8. Significance level

`alpha = 0.05`
`print("Significance Level:", alpha)`

Significance Level: 0.05

9. Perform two-proportion test

`conversions = [conversions_A, conversions_B]`
`users = [users_A, users_B]`
`z_stat, p_value = proportions_ztest(conversions, users)`

```
c:\Users\itlab\AppData\Local\Programs\Python\Py
prop = count * 1. / nobs
c:\Users\itlab\AppData\Local\Programs\Python\Py
p_pooled = np.sum(count) * 1. / np.sum(nobs)
c:\Users\itlab\AppData\Local\Programs\Python\Py
nobs_fact = np.sum(1. / nobs)
```

10. Calculate and display p-value

`print("\nZ-statistic:", z_stat)`
`print("P-value:", p_value)`

Z-statistic: nan
P-value: nan

11. Compare p-value with 0.05

`print("\nComparison of p-value with 0.05:")`

`if p_value < alpha:`
 `print("p-value < 0.05")`
`else:`

Conclusion:

The conversion rate of Group A and Group B was calculated and compared using a two-proportion z-test at a 5% significance level. Based on the p-value, the null hypothesis is not rejected if the p-value is greater than 0.05. Therefore, there is not enough statistical evidence to conclude that the conversion rates of Group A and Group B are significantly different.

`print("p-value >= 0.05")`

Comparison of p-value with 0.05:
p-value >= 0.05

12. Decision

`if p_value < alpha:`
 `print("Decision: Reject the Null Hypothesis (H0)")`
`else:`
 `print("Decision: Do not reject the Null Hypothesis (H0)")`

Decision: Do not reject the Null Hypothesis (H0)

13. Create bar chart

`groups = ["Group A", "Group B"]`
`conversion_rates = [rate_A * 100, rate_B * 100]`
`plt.bar(groups, conversion_rates)`
`plt.xlabel("Group")`
`plt.ylabel("Conversion Rate (%)")`
`plt.title("A/B Testing - Conversion Rate Comparison")`
`# Display values on bars`
`for i, rate in enumerate(conversion_rates):`
 `plt.text(i, rate, f"{rate:.2f}%", ha="center",`
 `va="bottom")`

`plt.show()`

